

MHKIE(G) interview drill

195 questions with model answers, one per page.

Cover the answer with your thumb, say your answer out loud, then scroll down to check.

Learn the reasoning, not the words: principle, Hong Kong application, then limitations.

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Tap a part to jump there, or use the PDF bookmarks.

Explain static liquefaction of a loose fill slope using a stress path in p' - q space. Why was recompaction of the top 3 m the traditional remedy?

- Loose fill is contractive. When sheared undrained (rapid rise in water from rainfall infiltration), pore pressure builds up instead of volume reducing.
- Stress path: the element starts with a high stress ratio under self-weight. On wetting, p' falls at roughly constant q , so the path moves left towards the instability line.
- At the instability line (peak of the undrained path, below the critical state line), a small disturbance causes a large pore pressure rise and strain-softening collapse to the very low steady state strength.
- Result: a sudden, highly mobile flow slide. Sau Mau Ping (1972, 1976) are the classic Hong Kong cases.
- Recomposition to $\geq 95\%$ maximum dry density makes the fill dilative: negative pore pressure under shear, so no brittle collapse. Top 3 m because historical failures were shallow, within the wetting front.

Assessor follow-up

Draw undrained paths of loose and dense sand on one p' - q plot; mark the critical state line and the instability line.

What is the design philosophy for soil nails in loose fill slopes (Geoguide 7 §5.8, GEO-HKIE 2011)?

- Soil nails do not prevent liquefaction. The design assumes the fill may liquefy and provides a nail-grillage system so the slope does not fail in a brittle, mobile way.
- Check three scenarios: (a) loose fill at its steady state undrained strength, minimum FOS 1.1 on any failure surface; (b) loose fill under drained conditions; (c) failure of the ground below the fill. Scenarios (b) and (c) use the normal FOS (Tables 5.4 to 5.6).
- Design strength: steady state strength c_{ss} . Without site-specific tests, a conservative $c_{ss} = 0.2 p'_{peak}$ may be used. Site-specific tests are strongly recommended.
- Nails bond into a competent stratum below the fill; the length within loose fill is treated as free length. Spacing not more than 2 m horizontally and 1.5 m vertically.
- Grillage: enough coverage to stop fill squeezing out, designed for bending, shear and (for hybrid layouts) axial tension, with nominal 0.3 m embedment against erosion.
- Hybrid layout: sub-horizontal nails in the upper part and steep nails in the lower part, with sub-horizontal nails about 40–50% of the total. No toe fixity is then needed. Other layouts need numerical analysis and an embedded concrete footing for toe fixity.

Assessor follow-up

Why is the c_{ss}/p' correlation not used for shear resistance at the base of the grillage? (Low stress level; Geoguide 7 suggests 3–10 kPa lower bound.)

What failure modes do you check in soil nail design, and what factors of safety apply (Geoguide 7)?

- External: overall stability, sliding, bearing.
- Internal, active zone: ground flowing around nails, bearing failure under heads, structural failure of the nail (tension, shear, bending), structural or connection failure of heads or facing, and local failure between heads.
- Internal, passive zone: pull-out at the soil-grout or grout-reinforcement interface.

Internal mode (Table 5.6)	Min FOS
Tensile failure of bar	$F_T = 1.5$
Pull-out, soil-grout	$F_{SG} = 1.5$ (transient loads, bonded in weathered granite or volcanic); 2.0 otherwise or for sustained loads
Pull-out, grout-reinforcement	$F_{GR} = 2.0$

Tensile capacity $T_T = f_y A' / F_T$. Soil-grout pull-out (effective stress method, Eq. 5.2):

$$T_{SG} = (c' P_c L + 2 D \sigma'_v \mu^* L) / F_{SG}$$

- P_c = grout sleeve perimeter, D = sleeve diameter, L = bond length in the passive zone, $\mu^* \approx \tan \phi'$.
- σ'_v is taken at mid-depth of the bond length, **capped at 300 kPa**.
- These factors are independent of the slope's consequence category.

Assessor follow-up

Why does F_{SG} differ between weathered granite and other soils? Why is a cap put on σ'_v ?

How do you decide corrosion protection for soil nails (Geoguide 7 §4.3 and §5.5)?

Step 1, site setting. "Aggressive" if affected by leakage from old developments, septic tanks, industry, livestock or cultivated land, or seepage from water mains or sewers, or near stray currents (substations, electrified rail, tramways). "Non-aggressive" if upslope is natural terrain with no signs of seepage. Otherwise "potentially aggressive", which needs a detailed assessment.

Step 2, marking scheme (Table 4.2): soil composition, resistivity, moisture content, groundwater, pH, soluble sulphate, made ground, chloride. Total mark: 0 non-aggressive; –1 to –4 mildly; –5 to –10 aggressive; –11 or below highly aggressive. pH below 4 or above 10 means aggressive regardless.

Step 3, protection class (all with hot-dip galvanising, minimum 610 g/m² zinc):

- Class 1: plus corrugated plastic sheathing.
- Class 2: plus 2 mm sacrificial thickness on the bar radius.
- Class 3: galvanising only.

Design life	Transient loads (Table 5.1)	Sustained loads (Table 5.9)
Up to 120 years	Class 1 for aggressive and highly aggressive; Class 2 for mildly and non-aggressive	Class 1 unless non-aggressive (then Class 2)
Up to 2 years	Class 3 (no aggressivity assessment needed)	

"Potentially aggressive" sites without assessment get Class 1 if design life exceeds 2 years.

Assessor follow-up

Why are fine-grained soils more corrosive than sands and gravels?

What is TDR testing of soil nails, and what are its limits?

- Time Domain Reflectometry sends an electrical pulse along a wire pre-installed on the bar. Bar and wire act as a transmission line; the reflection shows the bar length (and can indicate grout sleeve integrity).
- Geoguide 7 says TDR has proven reliability and is the simplest of the NDT methods examined; it is a useful audit tool, encouraging better workmanship.
- It is also used during pull-out test preparation to check the length of the grouted section.
- Limits: understand what it can and can't detect; it doesn't measure bond or capacity. Any new NDT method needs a known, consistent basis for interpretation (GEO TGN 18).

Bishop vs Janbu vs Morgenstern-Price: what are the assumptions, and which should you use for soil-nailed slopes?

Method	Equilibrium	Interslice forces	Surface
Bishop simplified	Moment (+ vertical force)	Normal only	Circular
Janbu simplified	Force only	Normal only, correction f_0	Non-circular
Morgenstern-Price	Force and moment	Normal and shear, assumed function	Any

- Geoguide 7 §5.3.4: methods satisfying force or moment equilibrium only may not give correct results, so **only methods satisfying both should be used** for soil-nailed systems (Shiu et al, 2007).
- All limit equilibrium methods assume rigid-plastic behaviour and one FOS along the surface: no deformations.
- Use finite element or finite difference analysis for steeply inclined nails, deformation checks, or sustained loads (e.g. strength reduction method).

Assessor follow-up

Why might a force-only method misjudge the effect of nail forces?

What FOS did you design to for a soil-nailed cut slope, and why?

New soil-nailed cut slopes (Table 5.4, ten-year return period rainfall):

Economic \ Life	Cat 1	Cat 2	Cat 3
A	1.4	1.4	1.4
B	1.4	1.2	1.2
C	1.4	1.2	> 1.0

A Category 1 slope also needs $FOS \geq 1.1$ for the worst predicted groundwater.

Existing cut slopes upgraded by nails (Table 5.5): 1.2 / 1.1 / > 1.0 for consequence-to-life Categories 1 / 2 / 3. Only valid with rigorous geological and geotechnical study (maintenance history, groundwater, rainfall and monitoring records), a long-standing slope, and loading, groundwater and slope form substantially unchanged. Otherwise use the new slope values.

Assessor follow-up

Give examples of Category 1 and Category 3 consequence-to-life (e.g. occupied buildings vs country parks, low-traffic roads).

How do you design a debris-resisting barrier for a natural terrain hazard?

- Natural terrain hazard study: desk study, aerial photo interpretation, field mapping, hazard types (open hillslope, channelised debris flow, boulder fall), design event volume, then debris mobility analysis for velocity and flow depth at the barrier.
- Rigid barrier impact by the hydrodynamic approach with dynamic coefficient α (GEO Report 270):

$$F = \alpha \rho v^2 h w$$

- Flexible barriers absorb energy through net deformation and dissipators. Check post-impact capacity, overflow and deflection clearance.

Assessor follow-up

Why does α vary? What decides rigid versus flexible?

How do you size slope surface drainage?

- Peak runoff by the Rational Method (Q in m^3/s , i in mm/hr , A in km^2):

$$Q = 0.278 C i A$$

- Use a high return period (typically 200 years for slope drainage, per GEO guidance) and a time of concentration suited to small steep catchments.
- Stepped channels on steep gradients to dissipate energy; catchpits at changes of direction. Allow for blockage; check freeboard and splashing at bends.

Assessor follow-up

What failures have been caused by drainage overflow, and what were the lessons?

Why is pit-by-pit excavation specified for works behind or below an existing retaining wall or slope?

- It limits the length of ground exposed at any time, so soil arching onto unexcavated ground keeps the wall or slope stable.
- Pits are dug in an alternating sequence (e.g. every third pit), and each is backfilled or concreted before neighbours are opened.
- Supervise pit width, depth, the sequence, time left open, and stop work before heavy rain.

Assessor follow-up

What would you do if a pit face started to ravel or water seeped in?

How do you upgrade an old masonry retaining wall with soil nails?

- Check sliding, overturning, bearing and overall stability, with nail forces as external loads (Geoguide 7 Fig. 5.7). Global FOS (Table 5.8): Set 1 sliding 1.5, overturning 2.0, bearing 3.0; Set 2 (rigorous study, long-standing wall, unchanged conditions) sliding 1.25, overturning 1.5. For masonry, the resultant must stay in the middle third.
- Fill behind old walls is uncertain, so bond nails only in a competent stratum (e.g. saprolite, dense colluvium) outside the active zone.
- Use permanent casing through the rubble zone behind the wall (prevents hole collapse, grout leakage, blocking drainage); ignore the cased length for pull-out.
- For loose-block masonry, use a concrete skin wall rather than isolated heads, embedded into the ground; or a grillage if the wall fabric must be preserved. Dowels fully into the masonry.
- Also repair leaking services and drainage; lesson from Kwun Lung Lau (1994).

What do raking drains do, and when should prescriptive subsurface drainage be added to a soil-nailed slope?

- Raking drains lower groundwater and relieve pressure; effectiveness depends on intercepting permeable paths and on maintenance.
- Geoguide 7 §5.10 recommends prescriptive subsurface drainage as a contingency where there is: seepage on the face; a permeability contrast that could cause perched water; a sizeable catchment; buried streams; high sensitivity to groundwater; nearby water-carrying services; or concern about closely-spaced nails damming groundwater.
- Especially relevant when hard cover is replaced by vegetation. Check that lowering groundwater won't harm nearby structures.

What do soil-nail heads and facing do, and how are heads sized?

- Heads give the reaction that mobilises tension in each nail, confine the ground near the surface (raising mean effective stress), prevent local failure between nails, and help redistribute forces.
- Nail tension increases with head size or facing coverage.
- Isolated heads for slopes steeper than 45° : Geoguide 7 Table 5.7 gives square head widths of 400, 600 or 800 mm depending on slope angle, c' , ϕ' and bar size, with **minimum thickness 250 mm**. Alternatively use the UK DOT (HA 68/94) method.
- On gentle slopes (e.g. natural hillsides), heads are less effective because strain is not normal to the head base; use enhanced details.
- A grillage gives better resistance to local surface failure than isolated heads.

How does nail inclination affect soil nail performance?

- Nails work best in tension, aligned with the direction of maximum tensile strain in the soil. Mobilising bending and shear needs about ten times more movement than mobilising tension.
- Nails are usually inclined downwards at **typically 5° to 20°** to allow gravity or low-pressure grouting.
- Effectiveness falls significantly as inclination increases; steep nails may go into compression and reduce the normal stress on the slip surface.
- If steep nails are needed (e.g. physical constraints), check them under combined tension, shear and bending using numerical analysis, and check the deformation needed to mobilise their force.

Describe the soil nail pull-out test procedure and acceptance criteria (Geoguide 7 §6.3.2).

- Purpose: verify the soil-grout bond assumed in design; also tests workmanship and buildability. Do tests before working nails, at locations of likely low resistance (weaker ground, high groundwater).
- Sacrificial test nails installed like working nails, but grouted only at the bottom: typically a **2 m** bond length, sealed off with a packer (TDR can confirm grouted length). Working nails must not be tested.
- Number: commonly **2% of working nails, minimum two**.
- Loading cycles: T_{DL1} (allowable pull-out of the 2 m bond), then $T_{DL2} = T_{DL1} \times F_{SG}$, each held 60 minutes; then up to the maximum test load, 90% of bar yield ($T_p = 0.9 f_y A$) unless ultimate bond is reached first. Use a larger bar if needed.
- Acceptance: movement between 6 and 60 minutes not more than **2 mm or 0.1% of the bond length**.
- Fully grout test holes afterwards.

Assessor follow-up

Why must working nails not be used for pull-out tests? (Grout sleeve damage, and bond in the active zone misleads.)

When and how would you use back analysis of a slope?

- When a slope has failed or shown distress: set $FOS = 1.0$ for the conditions at failure to estimate strength or groundwater.
- The solution is not unique: many combinations of c' , ϕ' and water level give $FOS = 1.0$. Fix the best-known values and solve for one unknown.
- Use it to check parameters from tests, not to replace them.

Why do most Hong Kong landslides happen in heavy rain, even on slopes without a high water table?

- Infiltration creates a wetting front that removes suction, so apparent cohesion disappears in the near-surface soil (Lumb's wetting band concept).
- Perched water can build up above less permeable layers.
- Short-duration, high-intensity rainfall drives many shallow failures; antecedent rain matters for deeper ones. The Landslip Warning system is based on rainfall-landslide correlations.

How do you assess and mitigate a boulder fall hazard?

- Assess: boulder size, how it is supported, undercutting, joint orientation, and the run-out path to facilities.
- Stabilise in place: dowels, buttresses, wire rope nets, or removal or splitting where safe.
- Protect: catch fences, flexible barriers or rigid barriers on the run-out path.

What are the benefits and risks of vegetation on slopes and walls?

- Benefits: erosion control, shallow root reinforcement, transpiration that lowers water content, and appearance.
- Not usually relied on in stability calculations, because the effect is shallow and variable.
- Risks: trees on walls can jack masonry apart, windthrow can pull out the surface, and roots can damage drains.

What are prescriptive measures, and when can soil nails be designed prescriptively?

- Pre-determined, experience-based, suitably conservative works to improve stability without detailed GI and design analysis. Used for preventive maintenance, urgent repairs and upgrading of existing soil cut slopes and retaining walls.
- Qualifying criteria cover type of works, slope or wall geometry, and consequence of failure: see GEO Publication No. 1/2009.
- No GI, so soil aggressivity is judged from site setting; "potentially aggressive" sites need Class 1 corrosion protection if design life exceeds 2 years.

What maintenance does a soil-nailed slope need (Geoguide 5, Geoguide 7 §7)?

- Routine Maintenance Inspections: at least yearly, by non-specialists, clearing drains and repairing surface cover.
- Engineer Inspections for Maintenance: **every 5 years for consequence-to-life Category 1 and 2 slopes, every 10 years for Category 3.** May be reduced if a post-construction review confirms satisfactory performance.
- Regular monitoring of buried water-carrying services that could affect the slope.
- Monitoring is not usually needed for nails carrying transient loads; it is for sustained loads.

What slope facing would you choose for a soil-nailed cut slope?

- Facings are soft (vegetation), flexible or hard. Flexible and hard facings can be structural, spreading load between heads and improving local stability.
- Vegetation is the first choice where safety and maintenance allow, but local washouts on vegetated soil-nailed slopes have occurred, often where hard cover was replaced by vegetation.
- On steep soil slopes, use a non-degradable erosion control mat with steel wire mesh, stretched and anchored to the heads. For high-consequence cases, consider an actively stressed mesh or grillage.
- Consider erosion control, local stability, construction, time to establish vegetation, maintenance, cost and appearance. Add buffer zones, catch fences or toe walls where the slope is steep and consequence is high.

A new building is planned close to the crest of an existing slope. What do you check?

- Add the foundation load as a surcharge in the slope stability analysis, and check the FOS still meets the required standard.
- Bearing capacity of the foundation is reduced near a slope crest; use sloping-ground factors or analyse the combined slip surface.
- Check construction effects: excavation for the foundation, vibration from piling, and changes to drainage or infiltration.
- Consider founding below the potential slip surface (e.g. piles) so no load reaches the slope.

How is debris mobility analysed for natural terrain hazards in Hong Kong?

- Debris is modelled as a continuum flowing down the terrain, giving velocity, flow depth and run-out distance (e.g. GEO's 3d-DMM).
- A common rheology is Voellmy: a friction term plus a turbulence term. Parameters are calibrated against back-analysed Hong Kong landslides.
- Results give the design impact velocity and thickness at the barrier, and whether debris reaches facilities.

Assessor follow-up

How sensitive is the run-out to the friction coefficient, and how did you choose it?

What load cases do you consider for a rigid debris-resisting barrier?

- Dynamic debris impact (hydrodynamic approach).
- Boulder impact within the debris, often with a cushion layer to spread the load.
- Static pressure from debris deposited behind the barrier, plus any water.
- Check structural capacity, sliding, overturning and bearing of the barrier; and overflow if the barrier is filled.

What are the main components of a flexible debris-resisting barrier?

- Ring or mesh net that catches debris, support and top ropes, posts, and energy-dissipating devices (brake elements) that absorb energy by deforming.
- Anchors take the rope forces into the ground; they need testing.
- Design checks: energy and force capacity, deflection and clearance downstream, post-impact capacity, and maintenance access for removing debris.

How do you set soil nail length, spacing and bar diameter?

- Length: enough bond length beyond the critical slip surface for the design force; check many surfaces, plus surfaces passing behind the nails. **Long nails (over about 20 m)** should be used with caution: harder to build, more likely to hit adverse ground, and need more movement to mobilise.
- Spacing: commonly **1.5 m to 2.0 m**, rows staggered for integral action. Too wide fails to prevent local instability; too close is costly and hard to install. Short intermediate nails can improve local stability.
- Bars: high yield deformed bars of **25, 32 or 40 mm**; small bars bend during installation of long nails.
- Practical limits: services, foundations, lot boundaries (need owners' permission beyond the lot).

A landslide has just occurred on a slope you are responsible for. How do you investigate it?

1. Safety first: keep people away, check for further movement.
2. Record quickly before evidence is lost: photos, dimensions, volume, scarp, debris path, seepage, damaged services.
3. Collect rainfall data, slope history, previous inspections and works.
4. Work out the mechanism and causes: geology, groundwater, drainage, services, maintenance.
5. Recommend urgent repair works and long-term upgrading.

What urgent measures would you take immediately after a landslide?

- Advise on evacuation or closure of affected areas; barricade the scar and debris path.
- Cover the scar with impermeable sheets, divert surface water away, check and isolate leaking services.
- Avoid loading the crest; remove debris only when safe.
- Monitor for further movement (survey markers, visual checks after rain).

What are the key design checks for a reinforced fill structure (Geoguide 6)?

- External: sliding, overturning (eccentricity), bearing capacity, and overall stability of the reinforced block.
- Internal: tensile rupture of reinforcement, pull-out, and connection strength at the facing.
- Compound: slip surfaces passing partly through the reinforced zone.
- Materials: well-graded fill with electrochemical limits for metal strips, and creep and damage reduction factors for polymer reinforcement.

What do you consider when designing a gabion wall?

- Acts as a mass gravity wall: check sliding, overturning, bearing and overall stability.
- Free-draining, so usually little water pressure, but a filter is needed behind to stop fines washing out.
- Durability: wire corrosion protection (galvanising, PVC coating) and durable rock fill of the right size.
- Tolerates movement, but not suitable where large settlement or high loads are expected.

What is quantitative risk assessment (QRA) for landslides?

- Risk = likelihood of landslide × probability it reaches people × vulnerability × number exposed.
- Individual risk: annual chance of death for the most exposed person. Societal risk: F-N curve of frequency against number of deaths.
- Results are compared with risk guidelines: unacceptable, tolerable if as low as reasonably practicable (ALARP), or broadly acceptable.
- Used for natural terrain and for prioritising works.

How would you design a cut slope where joints dip out of the face?

- Map the joints and do a kinematic analysis on a stereonet to find planar and wedge failure modes.
- Analyse critical blocks with joint water pressure.
- Options: flatten the cut to below the joint dip, rock bolts or dowels across the joints, buttresses, drainage holes, and protective netting or a catch fence.

Why does Geoguide 7 prefer soil nails to cutting back alone, even if both give the same calculated FOS?

- Robustness: GI can miss subtle adverse geological features. A nailed slope redistributes stress, overrides local weaknesses, and is less sensitive to undetected adverse ground and groundwater.
- Redundancy from many nails, and a ductile failure mode that gives warning.
- Construction merits: small plant for cramped sites, adjustable on site, less earthworks and tree felling.
- Limitations: buried services and obstructions, sterilised ground, lot boundaries, high groundwater, permeable ground causing grout loss, fines-rich ground prone to creep, deep-seated landslides, and deformation needed to mobilise force.

Compare drill-and-grout, self-drilling and driven soil nails.

- Drill-and-grout: most common. Gets through corestones, drilling spoil shows the ground, suits long nails, hole can be checked. But holes may collapse (casing needed) and drilling disturbs ground.
- Self-drilling: hollow bar drilled with a sacrificial bit, grout as flushing medium; fast and keeps holes open. Poor in corestones, alignment hard for long nails, and durability concerns (grout cover not assured without centralisers, coatings damaged).
- Driven: fast, little disruption, but only short nails, buckling in stiff ground or corestones, and corrosion-prone unless non-corrodible.

Explain the two-zone model and how a soil nail is mobilised.

- A potential failure surface separates the active zone (tending to detach) from the passive zone (stable). Nails tie the active zone to the passive zone.
- In the active zone, ground movement mobilises tension through nail-ground friction and the reaction at the head. The tension directly resists shear and increases normal stress on the failure surface.
- In the passive zone, the bond length resists pull-out through soil-grout and grout-bar bond.
- Maximum tension occurs near, but not exactly at, the failure surface. The model is a simplification for limit equilibrium; in reality there is a shear zone.

What is the presumed rock-grout bond for soil nails socketed into rock?

- Without detailed investigation, a presumed bond strength of **0.35 MPa** may be used for nails socketed into a partially weathered rock mass of PW90/100 or better.
- A **minimum rock socket of 2 m** is recommended, to allow for variable rock near the soil-rock interface.
- Higher values need justification by GI, testing and analysis.

Why is σ'_v limited to 300 kPa in the pull-out calculation, and what are the method's limits?

- The effective stress method ignores soil arching, restrained dilatancy, suction, hole roughness and over-break.
- At high overburden, arching around the hole may reduce contact stress more than dilatancy and roughness add, so σ'_v is capped at 300 kPa as a precaution.
- With the recommended F_{SG} , experience shows the method is adequately safe for usual Hong Kong conditions.
- Alternatives: SPT-N or pressuremeter correlations, or site-specific pull-out tests (Pun & Shiu, 2007).

How is the grout-reinforcement bond checked, and why is it rarely critical?

- Geoguide 7 Eq. 5.3: $T_{GR} = 2.25 \eta_1 \eta_2 f_{ctd} P_r L / F_{GR}$, using Eurocode 2 bond coefficients and grout tensile strength, with $F_{GR} = 2.0$.
- With deformed bars, bearing of the ribs on the grout gives most of the bond, so this interface is usually much stronger than the soil-grout interface.
- It can matter for short bond lengths, weak grout, or coated or sheathed bars (protection measures can affect bond).

What ground conditions cause buildability problems for soil nails, and how do you manage them?

- Excessive grout leakage in: coarse fill (boulders, cobbles), coarse colluvium, erosion pipes, material boundaries and corestone margins, open joints, faults, landslide scars and tension cracks, and drainage lines.
- Leaked grout can dam groundwater; install piezometers if this is a concern.
- Long nails (over about 20 m) more likely hit water and weak zones, causing hole collapse and grout loss.
- For long or closely-spaced nails or difficult ground, do a site trial before main works, possibly with pull-out tests; put details and contingencies in the contract.

How should soil nails be grouted, and what do you do about excessive grout take?

- Grout the same day as bar insertion. Grout through a pipe to the bottom of the hole until clean grout returns at the top.
- Keep a pressure head of about **1 m above the hole mouth** until initial set, to make up for leakage.
- Record grout volume for each hole. If grout take is excessive, notify the designer: options include staged grouting or casing over the leak, and review of the design model.
- Avoid grouting under water; provide dewatering (e.g. raking drains) if needed.
- With corrugated sheathing, prevent sheath distortion or floating during grouting.

What are the key supervision checks during soil nail construction?

- Bars and accessories: size, grade, length, galvanising and sheathing intact (no pitting), couplers tight, centralisers adequate.
- Drilling: position, length, inclination and bearing (check with protractor and compass; Eastman camera for long holes), ground and water found, hole left open only briefly, hole clean (mirror and light, or CCTV).
- Insertion: keep a minimum clearance at the hole bottom for grout cover; withdraw and clear if obstructed.
- Grouting and heads (see A41, A49); pull-out tests; excavation sequence.
- Full-time supervision, with insertion, grouting and pull-out tests individually checked. Geoguide 7 Fig. 6.1 gives a sample checklist.

Assessor follow-up

Which aspects can't be verified after installation, and so need the closest supervision?

When is a creep test needed, and what are its acceptance criteria?

- For nails carrying sustained loads and bonded in soil, to check susceptibility to long-term creep (especially fines-rich soil).
- Can be part of a pull-out test: one cycle to $T_c = T_{DL2}$ (allowable bond resistance $\times F_{SG}$), held 60 minutes.
- Acceptance: movement between 6 and 60 minutes not more than 2 mm or 0.1% of bond length, **and** the creep rate (movement per log time) decreasing throughout.
- If it fails, revise the design bond strength or move the bond zone to better ground, then re-test.

What is the difference between soil nails carrying transient and sustained loads, and why does it matter?

- Transient: the system is stable without nails most of the time; nails act mainly during e.g. high groundwater after heavy rain. Most Hong Kong slope upgrading is this case.
- Sustained: stability can't be maintained without the continued action of the nails.
- Sustained loads need: deformation analysis, creep tests when bonded in soil, stricter corrosion protection (Table 5.9), $F_{SG} = 2.0$, monitoring of movement and nail loads during construction (extended one wet season in unusual conditions), and a performance review before substantial completion.

What are the special considerations for soil nails in temporary excavations?

- Same FOS as permanent new slopes, but with groundwater and loading likely during the temporary works.
- Excavate in stages; the exposed height at each stage is set by its own stability.
- **Build heads and facing before the next stage.** Some large landslides have occurred in temporary cuts with nails but no heads.
- If nails carry sustained loads or movement could affect neighbours, do deformation analysis and monitoring. Take care with creep in cohesive soils.

What alternative soil nail reinforcement materials exist, and what are their drawbacks?

- Fibre reinforced polymer (carbon, glass, aramid): strong, light, corrosion resistant; but low bending and shear capacity and brittle failure, so a very high FOS may be needed.
- Stainless steel (austenitic, duplex) or stainless-clad bars: corrosion resistant and ductile; costly, little local experience.
- High tensile strand or bar: flexible for tight sites; corrosion-prone and nail-ground interaction not well understood.
- Check combined actions, strain compatibility, grout bond, ductility, durability, handling and testing needs.

When can a loose fill slope be upgraded by soil nailing rather than recompaction?

- Geoguide 7 §5.8.1 qualifying criteria: relative compaction of the fill **not less than 75%**; no significant distress, history of movement, heavy seepage or weak zone (e.g. silt or clay layer); and not in an old valley where groundwater may be high.
- Otherwise use other options such as recompaction.
- Fill formed by end-tipping without compaction is regarded as loose. Fill is loose if contractive or strain-softening; dense if properly compacted and dilative (then design as a cut slope).
- Expect construction difficulties: ground loss, hole collapse, grout loss. Limit open holes, use permanent casing or pre-grouting, and monitor crest facilities.

What could cause groundwater damming by soil nails, and how would you check for it?

- Normally nails and grout have no significant damming effect.
- Excessive grout leakage can seal permeable paths and raise groundwater behind the nailed zone.
- If concerned, install piezometers just behind the nailed zone and monitor before and after nailing, placing them where leaked grout is unlikely to block them.
- Prescriptive raking drains can be a contingency.

How should soil-nail heads be constructed to avoid defects?

- Honeycombing beneath the bearing plate is a common defect.
- With sprayed concrete, build in two stages: spray to a set thickness, then fix the bearing plate and tighten the nut, then spray the second stage.
- Clean the bar threads and protect them (galvanised or zinc-rich paint). Embed the corrugated sheathing into the head as specified.
- Uncover some heads as a quality check if needed. Recess heads and paint matt where appearance matters.

What tests are carried out on soil nail materials and installed nails?

- Material compliance: grout for crushing strength, bleeding and flow cone efflux time; bars for tensile, bend and re-bend tests (frequency per the General Specification for Civil Engineering Works).
- Pull-out tests on sacrificial nails (2% of working nails, minimum two); creep tests where needed.
- Non-destructive: TDR for bar length and grout integrity.
- Destructive (rare): stitch drilling, over-coring or excavation to exhume nails; hard to keep alignment.

State the principle of effective stress and explain why it matters.

- Terzaghi: $\sigma' = \sigma - u$. Strength and volume change depend on effective stress, because only stress carried at grain contacts mobilises friction and causes deformation.
- Rising water lowers σ' and strength (rainstorm failures); pumping lowers u and raises σ' (dewatering settlement).
- In unsaturated Hong Kong residual soils, suction adds apparent cohesion. It is lost when wetting fronts advance, so shallow failures occur without a main water table rise.

Draw the Mohr circle at failure for a drained triaxial test. Where is the failure plane?

- The circle passes through σ'_3 and σ'_1 and touches $\tau = c' + \sigma' \tan \phi'$.
- The failure plane is at $45^\circ + \phi'/2$ to the major principal plane (horizontal in standard compression).
- Pole method: draw from the pole parallel to any plane; the intersection with the circle gives the stresses on that plane.

Assessor follow-up

Why is the maximum shear stress plane (45°) not the failure plane?

Draw effective stress paths for: drained triaxial; undrained loose and dense soil; a foundation load; an excavation; a slope being wetted.

$$p' = (\sigma'_1 + 2\sigma'_3)/3, \quad q = \sigma_1 - \sigma_3$$

Case	Effective stress path
Drained triaxial compression	Straight line, slope 3
Undrained, loose / NC	Bends left: positive pore pressure
Undrained, dense / heavily OC	Bends right: negative pore pressure
Foundation load on NC clay (undrained)	q rises; effective path bends left
Excavation (lateral unloading)	q rises as σ_h falls, towards active failure
Slope wetting	Roughly constant q, p' falls: moves left

The gap between total and effective paths at any point is the excess pore pressure.

Explain dilation and contraction during shearing, in drained and undrained tests.

- Dense soil rides up over grains and dilates; loose soil collapses into voids and contracts.
- Drained: dense shows a peak then softens to critical state, volume increases. Loose shows no peak, volume decreases.
- Undrained: no volume change, so dilation gives negative pore pressure (strength up) and contraction gives positive pore pressure (strength down).
- Both end on the critical state line: constant volume and stress with continuing shear.

Assessor follow-up

Why is it unsafe to rely on peak strength in a large slope?

When do you use undrained (s_u) versus drained (c' , ϕ') parameters?

- Undrained when loading is fast compared with drainage: low-permeability clay during construction.
- Drained for granular soils and long-term conditions in clay.
- Cutting in clay: long-term is critical (pore pressures recover, strength falls). Embankment on soft clay: short-term is critical (strength rises with consolidation).

Explain Skempton's pore pressure parameters A and B.

$$\Delta u = B [\Delta \sigma_3 + A (\Delta \sigma_1 - \Delta \sigma_3)]$$

- B reflects saturation: near 1 when saturated (triaxial check, typically $B \geq 0.95$).
- A at failure is about 1 for NC clay and can be negative for heavily OC clay or dense sand.

What is arching? Give geotechnical examples.

- Stress transfer from a yielding part of the soil onto stiffer neighbours, through shear resistance.
- Examples: around tunnels, between stabilising piles in a slope, between nail heads behind a facing, trapdoor problems, piled embankments.
- It relies on shear strength, so it can be lost by saturation, vibration or excessive movement.

What is the critical state line, and how is M related to ϕ' ?

- The state where soil shears at constant volume and constant stress. In p' - q space it is $q = M p'$; in e - $\ln p'$ space it is parallel to the normal compression line.
- For triaxial compression:

$$M = 6 \sin \phi'_{cs} / (3 - \sin \phi'_{cs})$$

- Soil "wetter" than critical contracts; "drier" dilates.

What is K_0 , and how does overconsolidation change it?

- The ratio of horizontal to vertical effective stress with no lateral strain.
- Normally consolidated (Jaky): $K_0 \approx 1 - \sin \phi'$, typically 0.4–0.5.
- Overconsolidated: K_0 rises, roughly $K_0(\text{NC}) \times \text{OCR}^{\sin \phi'}$; can exceed 1 near the surface in heavily OC clay.

Explain Rankine active and passive earth pressure and their assumptions.

$$K_a = (1 - \sin \varphi') / (1 + \sin \varphi') \quad K_p = (1 + \sin \varphi') / (1 - \sin \varphi')$$

- Assumes a smooth vertical wall, horizontal ground and a full plastic state. Coulomb adds wall friction and sloping ground.
- Active pressure needs only small wall movement; passive needs much larger movement, so passive is usually factored down.
- Add water pressure separately, using effective stress for the soil part.

Write the bearing capacity equation and explain its terms.

$$q_{ult} = c'N_c + q N_q + \frac{1}{2} \gamma B N_\gamma$$

- Cohesion, surcharge (embedment) and self-weight terms; N factors depend on ϕ' . Add shape, depth and inclination factors.
- A water table at or near the base roughly halves γ in the last term.
- Failure modes: general shear (dense), local or punching shear (loose or soft).

What are the three components of settlement?

- Immediate (elastic, undrained distortion): $s = qB(1 - \nu^2)I/E$.
- Primary consolidation: from pore pressure dissipation, controlled by m_v or C_c and c_v .
- Secondary compression (creep): continues after pore pressure has dissipated, controlled by C_α ; important in marine mud and organic soils.

State Terzaghi's 1D consolidation time relationship and its key implication.

$$T_v = c_v t / H_{dr}^2$$

- H_{dr} is the longest drainage path: half the thickness if drained both sides.
- Time scales with H^2 , so halving the drainage path cuts time by four. That is why vertical drains work.
- Assumptions: 1D flow, saturated, small strain, constant c_v and k .

What is residual strength, and when is it relevant?

- The lowest strength after large shear displacement, when platy clay particles align along the shear surface.
- Relevant for reactivated landslides, old shear surfaces, and clay-infilled joints.
- Measured by ring shear or reversing shear box tests.

How does suction affect the strength of unsaturated soil?

$$\tau = c' + (\sigma - u_a) \tan \varphi' + (u_a - u_w) \tan \varphi^b$$

- Matric suction ($u_a - u_w$) adds apparent cohesion (Fredlund's extended Mohr-Coulomb).
- The soil-water characteristic curve links suction to water content.
- Not normally relied on in design because rain can remove it.

Explain the compaction curve and why fill is specified at 95% relative compaction.

- Dry density against water content peaks at the optimum moisture content (OMC) and maximum dry density (MDD); the zero air voids line bounds it.
- Dry of optimum: stiffer, more brittle, may collapse on wetting. Wet of optimum: more flexible, lower permeability.
- 95% relative compaction makes fill dense enough to be dilative, avoiding liquefaction and limiting settlement. Check by sand replacement or nuclear density tests.

What controls the shear strength of saprolite such as CDG?

- Typically low c' (around 0–10 kPa) and ϕ' around 35–38°; check against Geoguide 1 typical values and your own tests.
- Relict structure matters: relict joints, especially with kaolin infill, can be much weaker than the intact material.
- Suction adds strength in unsaturated conditions but is lost on wetting.
- Sample disturbance tends to underestimate strength from structure and bonding.

In an undrained triaxial test, how do the total and effective stress Mohr circles compare?

- They have the same diameter, because the deviator stress is the same.
- The effective circle is shifted left by the pore pressure u (right if u is negative).
- Undrained strength s_u is the radius of the total circle; the effective circle touches the c', ϕ' envelope.

Why is soil stiffness not a single number, and why does it matter?

- Stiffness falls with strain: very high at small strain (G_0 from seismic tests), much lower at larger strain.
- Typical strains around foundations and excavations are small, so lab stiffness from large-strain tests underestimates real stiffness.
- Using the wrong stiffness gives wrong movement predictions for excavations and foundations.

What causes overconsolidation?

- Removal of overburden by erosion or excavation.
- Desiccation (drying) near the surface.
- A fall then rise in groundwater level.
- Ageing and cementation, giving an apparent preconsolidation pressure.

What are the particle size boundaries for soil?

- Clay < 0.002 mm; silt 0.002–0.063 mm; sand 0.063–2 mm; gravel 2–63 mm; cobbles 63–200 mm; boulders > 200 mm.
- Description follows Geoguide 3 (and BS 5930): strength or density, colour, structure, then the main soil name.

What happens to clay when it is unloaded, for example at the base of an excavation?

- Unloading reduces total stress; pore pressure drops (can go negative), then the clay takes in water and swells over time.
- Effects: base heave, uplift on base slabs, tension in piles.
- Short-term the clay is stronger (negative pore pressure); long-term it softens, so the long-term condition is critical.

How do you estimate the shear strength of a rock joint?

- Clean rough joint: basic friction φ_b plus roughness. Patton: $\varphi_b + i$ (asperity angle).
- Barton: $\tau = \sigma'_n \tan[\varphi_r + \text{JRC} \log_{10}(\text{JCS}/\sigma'_n)]$, so roughness counts for less at higher normal stress.
- Infilled joint: if the infill is thicker than the asperities, use the infill strength, which can be much lower.

A long slope has a piezometer in it. Mark the piezometric level it will show.

- Assume steady seepage parallel to the slope: flow lines parallel, equipotentials perpendicular.
- Follow the equipotential through the tip up to the phreatic surface. The standpipe water rises to the elevation where that equipotential meets the phreatic surface, not to the water table directly above.
- So the reading is lower than the water table above the tip. At a point z below the phreatic surface (vertically), for slope angle β :

$$u = \gamma_w z \cos^2 \beta$$

Assessor follow-up

What if flow is not parallel, e.g. upward seepage at the toe?

Derive the FOS of an infinite slope with seepage parallel to the surface ($c' = 0$).

$$F = (\gamma' / \gamma_{sat}) \times (\tan \varphi' / \tan \beta)$$

γ' / γ_{sat} is about 0.5, so a fully saturated slope with surface-parallel seepage has roughly half the FOS of the dry slope. That is why rainfall is Hong Kong's main landslide trigger.

Derive the condition for hydraulic (base) failure due to upward seepage.

- Upward flow reduces σ' . Heave or boiling occurs when $\sigma' = 0$, at the critical gradient:

$$i_c = \gamma' / \gamma_w \approx 1.0$$

- $FOS = i_c / i_{exit}$. For sheet-pile cofferdams use Terzaghi's block method: submerged block weight (width $\approx$ half the embedment) versus uplift from excess pore pressure at its base.
- Remedies: deeper cut-off, filter/weighting blanket, relief wells, lowering the external head.

Assessor follow-up

How would a thin clay layer below the base change the check? (Uplift of the clay under artesian head.)

How do you draw a flow net and calculate seepage from it?

- Flow lines and equipotentials cross at right angles, forming curvilinear squares. Impermeable boundaries are flow lines; water-filled boundaries are equipotentials.
- Seepage per unit width:

$$q = k H N_f / N_d$$

- Anisotropic soil: scale horizontal distances by $\sqrt{(k_v/k_h)}$ and use $k' = \sqrt{(k_h k_v)}$.

How do you set the design groundwater level for a slope?

- Combine piezometer and Halcrow bucket records (peak levels), rainfall records and slope history; extrapolate to the 10-year return period from the response to known storms.
- Consider transient perched water above less permeable layers (colluvium over saprolite, relict joints) and leaking water mains.
- Instruments: Casagrande standpipes (slow response), vibrating wire piezometers with loggers (fast, for rainfall events), Halcrow buckets (maximum level between readings).

How do you measure permeability, and why do results vary?

- Lab: constant head (sands) and falling head (silts, clays).
- Field: falling or rising head tests in boreholes or standpipes, packer tests in rock, and pumping tests (most representative).
- Field values are usually higher, because they include fabric, joints and lenses that small samples miss.

What are the risks of dewatering for an excavation, and how do you control them?

- Lowering groundwater raises effective stress, causing settlement of nearby compressible ground and buildings on shallow foundations.
- It can also draw in fines, causing loss of ground.
- Controls: cut-off walls to reduce drawdown outside, recharge wells, monitoring of groundwater and settlement with trigger levels.

What is rapid drawdown, and why is it critical?

- When water outside a slope (reservoir, river) drops quickly, pore pressures inside the slope lag behind.
- The supporting water load is lost while internal pore pressure stays high, so FOS drops sharply.
- Critical for dams, catchwaters and slopes next to water. Analyse with undrained or transient seepage methods.

When would you use transient seepage analysis instead of a steady state assumption?

- When groundwater changes with time: rainfall infiltration, perched water build-up, rapid drawdown, dewatering.
- Uses unsaturated permeability functions and the soil-water characteristic curve, with a rainfall boundary (e.g. finite element software).
- Results are sensitive to input; calibrate against piezometer records where possible.

State the filter design criteria for a granular filter.

- Retention: $D_{15}(\text{filter}) / D_{85}(\text{soil}) \leq 4-5$, so soil particles don't wash through.
- Permeability: $D_{15}(\text{filter}) / D_{15}(\text{soil}) \geq 4-5$, so water passes freely.
- Also check uniformity and avoid segregation. Geotextiles use equivalent opening size criteria.

How do leaking water-carrying services affect slopes, and how are they controlled?

- Leaks add water locally, raising pore pressure and softening soil; a known cause of failures (e.g. Kwun Lung Lau, 1994).
- Controls: locate services in desk study, leak detection, regular monitoring under Geoguide 5, diversion or sleeving away from slopes, and owner coordination.

How do you design drainage behind a retaining wall?

- Provide a drainage layer or filter behind the wall, weepholes through the wall, and a collector at the base.
- The filter must stop fines washing out but let water through freely.
- If drainage could block, design for higher water pressure, or for full hydrostatic pressure in the worst case.
- Maintain weepholes: blocked weepholes are a common sign of trouble.

What are soil pipes, and why do they matter in Hong Kong slopes?

- Natural erosion tunnels in colluvium and saprolite that carry concentrated water flow.
- They can raise pore pressure locally or cause sudden erosion and collapse; several Hong Kong failures have been linked to them.
- Look for outflow points and voids during mapping; they are hard to find in drillholes. Design drainage and surface protection to cope.

How is water pressure included in a rock slope planar failure analysis?

- Water in a tension crack of depth z_w pushes the block: $V = \frac{1}{2} \gamma_w z_w^2$.
- Water along the sliding plane gives uplift U , reducing the normal force and friction.
- $FOS = [cA + (W \cos \psi - U - V \sin \psi) \tan \phi] / (W \sin \psi + V \cos \psi)$, with ψ the plane dip.
- Drainage holes reduce U and V and are very effective.

Why might piezometers in the same slope give very different readings?

- Different layers: perched water above a less permeable layer versus the main water table.
- Different response times: standpipes respond slowly; tips in low permeability soil lag.
- Local effects: soil pipes, leaking services, relict joints.
- Check installation details (response zone, seals) before trusting a reading.

How do you plan a GI for an LPMit Stage 3 study?

1. Desk study: GEO records (GIU, slope and landslide records, previous GI), geological maps, aerial photos, utility records.
2. Site mapping to build the engineering geological model: fill, colluvium, saprolite, rock, and water.
3. Set objectives: stratigraphy at critical sections, parameters, groundwater, buried features.
4. Methods: drillholes at crest and toe on the critical section, trial pits and slope stripping, in-situ density for fill, Mazier samples for triaxial tests, piezometers and Halcrow buckets.
5. Stay flexible: revise as results arrive.

Reference: Geoguide 2.

Assessor follow-up

What would make you add drillholes after the first results?

Describe the weathering grades of Hong Kong granite. How do you tell CDG from HDG?

- Grades I (fresh) to VI (residual soil), per Geoguide 3.
- Grade V (CDG): a soil, breaks down by hand, slakes in water, rock texture preserved.
- Grade IV (HDG): cannot be broken by hand, does not slake readily, breaks by hammer.
- Relict joints with kaolin or manganese-oxide infill can be weak planes. Corestones can give false rockhead readings.

Assessor follow-up

Typical friction angle of a kaolin-infilled joint versus intact CDG?

Describe the three stages of a CU triaxial test with pore pressure measurement.

1. Saturation: raise cell and back pressure together, small effective stress. Check B (typically ≥ 0.95).
2. Consolidation: apply target effective stress; record volume change against time. Time to 100% consolidation sets the shear rate.
3. Shearing: undrained, slow enough for pore pressure to equalise. Measure deviator stress and pore pressure; plot effective stress paths.

CU tests are common because they are faster than drained tests and still give effective stress parameters.

Assessor follow-up

How can you tell from the stress path whether a sample was dilative or contractive?

When would you specify a shear box test instead of a triaxial test?

- Shear box: shearing along a defined plane (relict joint, soil-structure interface) and residual strength by reversals. Cheap and quick.
- Limits: forced failure plane, no pore pressure control or measurement, non-uniform stress.
- Triaxial: drainage control and pore pressure, so you can follow the stress path.

How do you derive design c' and ϕ' from triaxial results for a shallow slope?

- Plot in $s'-t$ or $p'-q$ and fit an envelope over the stress range on the slip surface: low stress for shallow slopes.
- Envelopes are often curved. A straight fit over high stresses overestimates c' at low stress, which is unconservative.
- Use a cautious estimate, not the mean; check against back analysis or GEO typical values.

What are the limitations of SPT in Hong Kong ground?

- N depends on hammer energy, borehole disturbance (base heave in loose sand below water) and corestones or gravel.
- In saprolite, N varies widely with relict structure. Use it as an index, not a direct strength value, unless locally correlated.

What do you get from an oedometer test, and how do you use it?

- e-log σ' curve, C_c , C_r , preconsolidation pressure σ'_p (Casagrande), m_v , c_v .
- Uses: settlement magnitude (m_v or C_c , C_r) and timing (c_v), e.g. reclamation and surcharge design.
- Lab c_v often underestimates field c_v because it misses fabric such as sand lenses. Use piezocone dissipation and settlement monitoring.

What is a Mazier sampler, and why is it used in Hong Kong?

- A triple-tube core barrel with a spring-loaded inner tube that projects ahead of the bit, protecting the sample from drilling fluid.
- It recovers good-quality samples of weathered soils (CDG, CDV) that are too hard for push samplers.
- Samples are still disturbed somewhat; check for drilling damage before testing.

Define TCR, SCR and RQD.

- TCR: total core recovered as a percentage of the run length.
- SCR: solid full-diameter core as a percentage of the run.
- RQD: total length of sound core pieces ≥ 100 mm, as a percentage of the run. An index of rock mass fracturing, used in Q and RMR.

What do you record in a rock joint survey?

- Orientation (dip and dip direction), spacing, persistence, roughness (JRC), aperture, infill type and thickness, wall strength, and water.
- Scanline or window mapping to avoid orientation bias; plot on a stereonet.
- Infill and persistence often control stability more than orientation alone.

What geophysical methods are used in GI, and what are their limits?

- Seismic refraction: rockhead profile; fails where a fast layer overlies a slow one.
- Resistivity: water, clay, cavities in marble; non-unique interpretation.
- Ground penetrating radar: shallow features, voids, buried services; poor in clay or wet ground.
- Always calibrate against drillholes.

What instruments would you specify to monitor a deep excavation?

- Inclinometers in or behind the wall, settlement markers on ground and buildings, tilt markers, piezometers and standpipes, strut load or strain gauges, utility monitoring.
- Set Alert-Alarm-Action levels from the predicted movements, with agreed responses at each level.
- Take baselines before work starts.

How do you investigate whether a fill slope is loose?

- Desk study: when and how the fill was placed (aerial photos, records).
- In-situ density tests (e.g. sand replacement) in trial pits at several depths, and lab compaction tests for maximum dry density, giving relative compaction.
- Also check fill thickness, groundwater and what's at the toe; triaxial tests can show contractive behaviour.

What index tests are done on soils, and what are they used for?

- Moisture content, particle size distribution, Atterberg limits, particle density, bulk density.
- Uses: classification, correlation of behaviour (e.g. plasticity with compressibility), filter design, compaction control.

What do you check when supervising a GI contract?

- Hole locations and levels, method compliance with the specification, sampling and SPT procedure, water level records at start and end of shift.
- Logging against Geoguide 3, sample handling and labelling, core boxes photographed.
- Safety: rigs on slopes, platforms, traffic, utilities. Update the GI as findings come in.

Assessor follow-up

Give an example where your checking found a logging or procedure error.

What does a piezocone test (CPTu) give, and where is it useful in Hong Kong?

- Cone resistance, sleeve friction and pore pressure with depth: continuous soil profiling.
- Undrained strength from $s_u = (q_t - \sigma_v)/N_{kt}$; consolidation coefficient from dissipation tests.
- Very useful in marine deposits; limited in residual soils with corestones, boulders or dense layers.

What is a field vane test, and what correction is applied?

- A four-bladed vane is pushed into soft clay and rotated; torque at failure gives undrained strength.
- Vane strength often overestimates field strength in high-plasticity clay, so a correction based on plasticity index is applied (Bjerrum).
- Only for soft to firm clays without sand or shells.

How is the point load test used to estimate rock strength?

- A core or lump is loaded between two cones to failure; the result is corrected to a 50 mm size: $I_{s(50)}$.
- UCS is roughly $20\text{--}25 \times I_{s(50)}$, but the factor varies with rock type, so calibrate with some UCS tests.
- Quick and cheap; useful for many readings along core.

What does a pressuremeter test measure?

- A probe in a borehole expands against the wall; pressure against expansion gives the shear modulus and limit pressure.
- Gives in-situ stiffness for excavation and foundation analysis, including unload-reload stiffness.
- Results depend on borehole quality and installation disturbance.

How does rock strength change with weathering grade in Hong Kong granite?

- Fresh granite (Grade I-II) is very strong, often over 100 MPa UCS.
- Strength drops sharply through Grade III to Grade IV; Grade V and VI behave as soil.
- Weathering is uneven, so rockhead can be irregular with corestones; drilling must go far enough into sound rock to confirm.

What is a plate load test, and what are its limits?

- A rigid plate on the ground is loaded in stages; settlement gives bearing behaviour and stiffness of near-surface ground.
- Limit: it only tests ground to about $1.5\text{--}2 \times$ plate width deep, so a full-size foundation stresses deeper layers that the test misses.

What is a packer (Lugeon) test?

- A section of borehole in rock is sealed with packers and water is pumped in at set pressures to measure intake.
- 1 Lugeon = 1 litre per minute per metre at 1 MPa. Gives rock mass permeability.
- The pressure-flow pattern shows joint behaviour (e.g. washing out or jacking open).

Outline the CIRIA C580 approach to embedded retaining wall design.

- Serviceability: moderately conservative parameters, unfactored, to predict wall movement and ground settlement.
- Ultimate: partial factors on soil strength (or worst credible parameters) to check embedment, bending, prop loads and overall stability.
- Unplanned excavation allowance (10% of retained height, up to 0.5 m) and a minimum surcharge.
- Used in Hong Kong where deformation matters, e.g. near MTR structures.

Assessor follow-up

Why is C580 better for deformation-sensitive sites than a global factor of safety approach?

How do you calculate the overall stiffness of a strut system for a computer model?

- Series (strut, waling, connection: same force), compliances add:

$$1/k_{total} = 1/k_1 + 1/k_2 + 1/k_3$$

- Parallel (same displacement): $k = k_1 + k_2$.
- Axial stiffness per metre run = $EA/(L \cdot s)$, s = strut spacing. Symmetric excavation: centreline doesn't move, so use half the strut length.
- Reduce for connection slack, temperature and preload; check sensitivity.

What is the hoop stress in a circular excavation, and what are the risks?

- Ring compression $N = p \cdot r$ per unit height, so hoop stress $\sigma = p \cdot r / t$.
- Advantage: self-supporting in compression, no struts, open working space.
- Risks: out-of-roundness or uneven pressure causes bending; diaphragm wall panel verticality and joints are critical.

Explain negative skin friction and how you design for it.

- Cause: soil settles more than the pile (new fill on soft clay, dewatering), dragging it down above the neutral plane.
- Design: structural check for permanent load plus drag load; check downdrag settlement. Shaft resistance above the neutral plane doesn't count as capacity.
- Mitigation: bitumen slip coating, sleeves, surcharge and wait before piling, larger sections.

Assessor follow-up

Where is the neutral plane for an end-bearing pile on rock versus a friction pile?

What special investigation and design considerations apply to marble sites in Scheduled Areas?

- Some Scheduled Areas are underlain by marble with solution cavities; geotechnical works need specialist control and GEO checking.
- Extensive GI, often drillholes at each pile, to locate cavities and classify marble by Marble Quality Designation (MQD).
- Options: found piles on sound marble of adequate thickness below the toe, grout cavities, or rafts in some cases. Watch karst features at rock surface.

Assessor follow-up

What other geotechnical controls apply in Scheduled Areas and Mid-Levels, and why?

Explain pile set-up and relaxation, and how dynamic testing is used.

- Set-up: capacity rises with time after driving as excess pore pressure dissipates and soil ages.
- Relaxation: capacity falls with time, e.g. in dense dilative sand or some weathered rocks.
- PDA with signal matching (CAPWAP) estimates mobilised capacity on restrike. Hong Kong also uses the final set (Hiley formula) for driven H-piles.

What are typical properties of Hong Kong marine mud, and how do you improve it?

- Very soft to soft, moisture content often 50–100%, s_u/σ'_v about 0.2–0.3 when NC, low c_v , significant creep.
- Improvement: PVDs plus surcharge (radial consolidation, Barron/Hansbo), deep cement mixing (airport third runway), stone columns, dredge and replace.
- Monitor with settlement plates, piezometers and inclinometers to control fill rate and confirm consolidation before surcharge removal.

Assessor follow-up

Why does lab c_v underestimate field rate, and what is smear around drains?

How do you use a stereonet to assess rock slope failure modes?

- Planar: joint dips out of the face, daylights ($\text{dip} < \text{slope angle}$), steeper than φ , strike within about $\pm 20^\circ$ of the face.
- Wedge: intersection line plunges out of the face, less steep than the face, steeper than φ .
- Toppling: steep joints dipping into the slope, striking roughly parallel to the face.
- Tunnel roof: check wedges free to fall under gravity and wedges sliding on one plane (e.g. Unwedge).
- Clay infill can lower φ well below clean joints.

How do you assess settlement from TBM tunnelling and set face pressure?

- Gaussian trough, $i = K \cdot z_0$ ($K \approx 0.5$ clay, 0.25–0.35 sand):

$$S(x) = S_{\max} \exp(-x^2 / 2i^2)$$

- Volume loss: $V_s = \sqrt{(2\pi)} \cdot i \cdot S_{\max}$.
- Face pressure: high enough to prevent collapse and limit volume loss, low enough to avoid blow-out at shallow cover.
- Building damage: greenfield settlement then Burland-type categories; monitoring with Alert-Alarm-Action levels.

Name some special Hong Kong rock types and their engineering significance.

- Eutaxite: welded tuff with flattened fiamme; can be anisotropic.
- Andesite and other volcanics: generally strong, can weather irregularly.
- Monzonite and syenite: intrusives that weather differently from granite.
- Marble: solution cavities.
- Know the geology under your own LPMit features from the HKGS 1:20,000 maps.

How do you choose between bored piles, driven H-piles, socketed H-piles and mini-piles in Hong Kong?

- Bored piles: large loads, can be founded on rock; slower, need base cleaning and inspection.
- Driven H-piles: quick, cost-effective in suitable soils; noise and vibration, limited by boulders.
- Socketed H-piles: steel H-section grouted into rock sockets; good where rockhead is shallow to moderate.
- Mini-piles: restricted access, underpinning.
- Factors: ground profile, rockhead depth, load, noise, vibration, nearby structures, cost and programme.

How is a static pile load test carried out and judged?

- Maintained load test, typically to twice working load, applied in stages with holding periods; record settlement against load and time.
- Acceptance in Hong Kong is by settlement criteria in the Code of Practice for Foundations, based on elastic shortening plus a toe allowance. Know the formula.
- Also check residual settlement after unloading.

How are rock socketed piles designed?

- Capacity from end bearing and shaft friction in the rock socket. The Code of Practice for Foundations gives presumed values by rock grade and quality.
- Confirm rock quality below the base (e.g. pre-drilling a few metres below founding level).
- Base cleanliness and socket inspection are critical; soft debris at the base cuts capacity.

Name ground improvement methods and when each fits.

- Vertical drains plus surcharge: soft clay consolidation.
- Deep cement mixing: soft clay strength and settlement control.
- Jet grouting: cut-offs, underpinning, base plugs, mixed ground.
- Permeation grouting: sand and gravel cut-offs.
- Vibro-compaction and stone columns: loose sand, reclamation fill.
- Dynamic compaction: loose granular fill, away from sensitive structures.

Compare excavation support wall types.

- Sheet piles: quick, watertight; limited by boulders and depth, noisy if driven.
- Soldier piles and lagging: cheap, not watertight.
- Pipe pile walls: through mixed ground with boulders, used in Hong Kong.
- Contiguous or secant bored piles: stiffer; secant walls cut off water.
- Diaphragm walls: deep, stiff, watertight; can be permanent walls.
- Choose by depth, groundwater, ground, movement limits and space.

How do you check basal heave in a soft clay excavation?

$$FOS = N_c s_u / (\gamma H + q)$$

- Bjerrum and Eide: N_c depends on excavation shape and depth.
- Remedies: deeper embedment into stiffer soil, base improvement (jet grout or DCM slab), excavating in strips under water.

How do you estimate ground movement behind an excavation, and its effect on buildings?

- Empirical charts (Peck, Clough and O'Rourke) relate settlement to depth and support stiffness; numerical analysis for detail.
- Movements depend on wall stiffness, prop spacing and preload, workmanship and dewatering.
- Assess buildings with damage categories (Burland), and set monitoring trigger levels.

What measures are used to stabilise rock slopes?

- Scaling of loose blocks, rock bolts and dowels, buttresses for overhangs, mesh and netting, shotcrete in weak zones, and drainage holes.
- Protective measures: catch fences, barriers, ditches at the toe.
- Choose from kinematic analysis and block size.

What are the Q and RMR rock mass classification systems used for?

- Q: RQD, joint set number, roughness, alteration, water and stress reduction factor; widely used for tunnel support charts.
- RMR: strength, RQD, spacing, joint condition, water, orientation adjustment.
- Both are empirical; use for preliminary support and correlations, then verify by observation.

What is the Hoek-Brown criterion, and why use it?

- A non-linear strength envelope for rock masses, using intact strength σ_{ci} , material constant m_i , and the Geological Strength Index (GSI) to reduce to mass strength.
- Suits fractured rock masses, where a linear Mohr-Coulomb fit is poor.
- Not for rock controlled by a few dominant joints; use kinematic analysis then.

What are the geotechnical concerns with drill-and-blast tunnelling in Hong Kong?

- Vibration and airblast: peak particle velocity limits for nearby structures, slopes and utilities; blast design and monitoring.
- Explosives need statutory approval and control.
- Overbreak, wedge stability, groundwater inflow and drawdown; face mapping to confirm support.

What are the main geotechnical issues in reclamation?

- Consolidation of underlying marine mud: settlement magnitude and timing, often handled by drains plus surcharge or DCM.
- Residual settlement after handover, affecting buildings, roads and utilities.
- Seawall stability, mud waves during filling, and liquefaction potential of loose sand fill.

How does a pile group behave differently from a single pile?

- Stress zones overlap, so group settlement is larger than single-pile settlement under the same load per pile.
- In clay, check block failure of the whole group as one unit.
- Capacity per pile in a group can be less (efficiency below 1) in clay, or similar in sand where driving densifies the ground.

How do you design piles for lateral load?

- Short rigid piles rotate and fail by soil yield; long flexible piles fail by bending in the pile (Broms' method).
- Load-deflection with p-y curves or a subgrade reaction model, or numerical analysis.
- Head fixity (free or fixed to a pile cap) strongly affects moments and deflection.

Compare prestressed ground anchors with soil nails.

- Anchors: active, prestressed and locked off; have a free length and a fixed (bonded) length; each anchor is proof tested.
- Nails: passive, fully bonded, take load only as the ground moves; tested by sampling.
- Anchors control movement better and suit high loads; they need durable corrosion protection and long-term monitoring of load.

What is the role of bentonite slurry in diaphragm wall construction?

- The slurry's pressure supports the trench sides; a filter cake forms on the trench wall so the pressure acts on the soil.
- Control density, viscosity, sand content and pH; desand before concreting.
- Keep the slurry level well above groundwater; a sudden loss of slurry can cause trench collapse.

What special controls apply to works near railway structures in Hong Kong?

- Railway protection areas: works need the railway operator's agreement and BD requirements (see the relevant Practice Note for railway protection).
- Strict limits on movement, vibration and loading of tunnels and structures; condition surveys before and after.
- Monitoring with Alert-Alarm-Action levels and agreed responses.

What methods are used to underpin existing foundations?

- Mini-piles through or beside the footing, needling beams.
- Mass concrete underpinning in short pits (like pit-by-pit).
- Jet grouting or permeation grouting to strengthen the ground.
- Monitor the building throughout; control sequence to limit movement.

What is the observational method, and when is it appropriate?

- Peck: design for the most probable conditions, plan contingency measures for worse conditions, monitor, and act on pre-set trigger values.
- Suits works where ground behaviour is uncertain and changes can be made during construction (excavations, tunnels).
- Not suitable where failure would be sudden with no warning, or where contingencies can't be applied in time.

How is tunnel support designed in sequential excavation (NATM-type) methods?

- Excavate in stages and install support (shotcrete, rock bolts, lattice girders) based on the rock or ground class found.
- The ground-support interaction: let the ground deform a little to mobilise its own strength, but not too much.
- Convergence and face monitoring confirm the support works; adjust the support class as needed.

What are the geotechnical considerations for rock caverns?

- Rock mass quality and joint orientation relative to the cavern axis; avoid major faults.
- Span and shape: wider spans need better rock and more support.
- In-situ stress, groundwater inflow and drawdown, and support design (bolts, shotcrete).
- Hong Kong has used caverns to free land; good rock is often available.

What are the problems with founding on fill?

- Fill may be loose, variable, and contain voids or organic matter: large and uneven settlement.
- Fill can keep settling under its own weight, and collapse on wetting if placed dry.
- Options: remove and replace, compact, ground improvement, or piles through fill (allow for negative skin friction).

(C8) A contractor offers you a gift, or presses you to certify unfinished work. What do you do?

- Refuse; never certify what you haven't verified. Accepting advantages can breach the Prevention of Bribery Ordinance (Cap. 201) and employer rules.
- Record and report to your supervisor or the project manager; follow the declaration procedure.
- Link to the HKIE Rules of Conduct: duty to the public and employer, integrity, working within competence.

Assessor follow-up

What if your own senior asks you to sign calculations you haven't checked?

(C2) Explain how slope safety responsibility works for a private slope in Hong Kong.

- Owners maintain slopes on their land; responsibility is shown in the Lands Department's SMRIS.
- If a private slope is dangerous or liable to become so, BD can serve a Dangerous Hillside Order under s.27A of the Buildings Ordinance. Your Stage 2 recommendations feed into this.
- Government slopes are upgraded under LPMit; works are checked by GEO.

Assessor follow-up

What happens if the owner does not comply with a DH Order?

(C7, C9) What are the main safety hazards on slope upgrading works, and how did you control them?

- Hazards: work at height on steep slopes, drill rigs on scaffolding, lifting, unsupported excavation (hence pit-by-pit), debris falling onto roads below, work in rainstorms.
- Controls: method statements and risk assessments, scaffold inspections, permit-to-work, rainstorm suspension, traffic management and hoarding, under Cap. 59.
- Give a real example from GE/2020/05 where you stopped or changed an activity.

(C9) How do you address sustainability in slope works?

- Prefer greening and soft landscape over hard cover; keep existing trees.
- Choose lower embodied carbon where equally effective (e.g. flexible vs large rigid barrier).
- Reuse excavated material; minimise waste; control noise, dust and runoff.
- Life-cycle thinking: maintenance access and durability reduce future resource use.

(C6, C7) Under NEC4, how are unforeseen ground conditions handled?

- Physical conditions are a compensation event (clause 60.1(12)) if an experienced contractor would have judged, at the contract date, that they had such a small chance of occurring that allowing for them was unreasonable.
- Early warnings and the risk register should surface ground risks early, e.g. targeted GI before main works.
- Use a real CE you assessed: entitlement, time and cost.

(C10) A resident refuses land entry for GI or works. How do you handle it?

- Explain purpose, safety benefit, duration and site arrangements in plain language. Listen to concerns (noise, privacy, damage) and offer practical measures.
- Involve the right departments, keep records, escalate through proper channels.
- Use your own example from the GI contracts.

(C1, C11) What are the key lessons from major Hong Kong landslides?

Event	Lesson
Sau Mau Ping (1972, 1976)	Loose fill liquefaction; led to GCO (now GEO) in 1977 and recompaction.
Po Shan Road (1972)	Groundwater in deep weathering profiles can trigger large failures.
Kwun Lung Lau (1994)	Leaking services behind old masonry walls.
Fei Tsui Road (1995)	Weak kaolin-rich layers; importance of the geological model.
Sept 2023 black rainstorm	Extreme rainfall, many landslides; climate resilience.

Check details against GEO reports before the interview.

(C11) How do you keep your knowledge current? Name a recent development.

- Give your actual CPD: HKIE Geotechnical Division seminars, GEO publications and TGNs, courses, papers.
- Prepare: smart slope monitoring and early warning, digital tools in LPMit, debris barrier design advances, climate change in design rainfall.
- Have two or three papers you can discuss.

What is your weakest geotechnical area, and what are you doing about it?

- Be honest: likely foundations, deep excavation and tunnelling.
- Show the plan: reading, CPD, exposure through colleagues or site visits.
- This shows self-reflection, which the interview explicitly assesses.

(C12) Give an example where site conditions differed from the design, and what you did.

- Use a real case from GE/2020/05 (e.g. rockhead or fill different from the GI, groundwater found, a buried feature).
- Structure: what you saw, how you judged the risk, who you informed, the change you made or proposed, and the outcome.
- Show that you recognised when to stop and get the designer's input.

(C6) How did you plan and control programme on your contract?

- Explain the programme reviews, critical activities, rain weather allowances, and how you tracked progress against Task Orders.
- Show an example of a delay you anticipated and the early warning or action you took.

(C6) How did you prepare budgets and expenditure forecasts?

- Cost estimates from quantities and rates (BQ), contingency for ground risk, year-by-year expenditure forecasts against the approved budget.
- Monthly tracking of interim payments, CEs and remaining risks; explain variances.
- Give real figures from your contract.

(C2) How is geotechnical control applied to public and private works in Hong Kong?

- Public works: GEO checks geotechnical submissions for slopes and retaining walls under the relevant Works technical circulars.
- Private works: Buildings Ordinance submissions to BD, with GEO advising on geotechnical aspects; RGE involvement where required.
- Special controls in Scheduled Areas and near railways.

Assessor follow-up

What is a Registered Geotechnical Engineer responsible for under the Buildings Ordinance?

(C7) How is site supervision organised for geotechnical works under the Buildings Ordinance?

- Code of Practice for Site Supervision: supervision plans, with Technically Competent Persons (TCP grades) assigned according to the works' risk.
- The RGE's stream supervises geotechnical works such as slopes, retaining walls and excavations.
- Relate to your own site supervision role and what you checked.

(C8) The client wants to cut cost by reducing the works below what you consider safe. What do you do?

- Explain the risk clearly, with evidence: what could fail and the consequences.
- Offer alternatives that save cost without lowering safety (e.g. a different nail pattern, staged works).
- Do not sign off a design you believe is unsafe; record the advice and escalate. Public safety comes first under the HKIE Rules of Conduct.

(C10) How would you explain landslide risk to residents or a District Council?

- Plain language: what the hazard is, how likely, what is at risk, what is being done, and how long it takes.
- Use photos and simple sketches, not jargon or FOS values.
- Be honest about uncertainty, and say what residents can do (e.g. reporting signs of distress).

(C6) How do you control quality on geotechnical works?

- Inspection and test plans with hold points (e.g. drilled length checked before grouting, rock exposed before concreting).
- Material tests (grout cubes, bar certificates), TDR or pull-out tests, records and photos.
- Non-conformance reports, corrective action, and checking the fix.

(C9) How should designers respond to climate change and more extreme rainfall?

- Use updated rainfall data and consider higher intensities in drainage design and design groundwater.
- Favour robust, redundant designs (e.g. hybrid nails, larger drainage capacity), and resilience against debris.
- Better monitoring and early warning; low-carbon solutions in construction.

Why are you applying for MHKIE(G), and what is your plan after?

- Link to your geotechnical career: you already take responsibility for slope design and site supervision.
- Route: MHKIE(G), then RPE(G), then RGE experience.
- Show commitment: CPD, Geotechnical Division activities, mentoring juniors.

Tell us about a mistake you made and what you learned.

- Choose a real, moderate mistake (e.g. a missed check or an assumption that proved wrong), not a disaster or a fake one.
- Explain how you found it, how you fixed it and who you told.
- End with the change you made to how you work. Assessors look for honesty and reflection.

(C10) How did you coordinate with other disciplines and parties on your projects?

- Name the parties: utility companies, traffic authorities, landscape architects and arborists, private owners, GEO, BD.
- Give an example of a conflict (e.g. nails clashing with services, tree preservation versus works) and how you resolved it.
- Show meeting records, clear drawings and early warnings.

(C11) What digital tools are changing geotechnical practice?

- LiDAR and drone surveys for terrain and slope mapping.
- Real-time monitoring with sensors and automatic alerts.
- BIM and 3D geological models for coordination and records.
- Be specific about what you have used yourself, and the limits (data quality, false alarms).

(C6) Compare a traditional contract (GCC) with NEC4.

- Traditional: Engineer administers, risk allocation fixed, claims often settled late and can be adversarial.
- NEC4: mutual trust and cooperation, early warnings and risk reduction meetings, compensation events assessed with time limits, an accepted programme as a key tool.
- Give your own NEC4 experience as evidence.

(C8) What would you do if you had a conflict of interest?

- Declare it to your employer or client as soon as you know.
- Step back from decisions affected by it unless the client agrees with full knowledge.
- Link to the HKIE Rules of Conduct and your company's procedures.

(C9) A serious accident happens on your site. What do you do?

- Make the site safe, get medical help, and stop the related work.
- Report to the right people and authorities as required, and preserve evidence.
- Investigate root causes (not just blame), revise method statements and risk assessments, and share lessons.

(C6) How do you supervise and develop junior staff?

- Give clear tasks, check their work, explain why, not just what.
- Let them take responsibility gradually, with review.
- Give an example: a site inspector or graduate you guided.

(C9) How can carbon be reduced in geotechnical works?

- Use less material: optimise design, avoid over-conservatism, reuse existing structures.
- Lower-carbon materials: cement replacement in concrete and grout, recycled aggregates.
- Choose low-carbon solutions (e.g. soil nails and greening instead of large concrete walls, flexible barriers).
- Reduce transport and waste: reuse excavated material on site.

What does Geoguide 1 cover, and what does it not cover?

- Recommended good practice for **new permanent earth retaining walls on land** in Hong Kong. It adopts the limit state method with partial factors (2nd edition, 1993; continuously updated e-version on the CEDD website).
- Wall types covered: gravity walls (mass concrete, crib, gabion, reinforced fill), reinforced concrete walls (L-shaped, inverted T, counterfort, buttressed), and cantilevered walls (e.g. bored pile walls).
- Not covered: maritime structures, assessment of existing walls, masonry walls, and temporary deep excavation support (sheet pile, soldier pile, diaphragm walls). For excavations, see GCO Publication No. 1/90, and now GEO Publication No. 1/2023.

Assessor follow-up

So which documents would you use to assess an old masonry wall? (e.g. Geotechnical Manual for Slopes, GEO Report on old masonry walls, Geoguide 7 if nailing.)

Explain the limit state approach in Geoguide 1.

- A limit state is reached when the wall, or part of it, fails to meet a performance criterion. Each limit state is checked separately and shown to be sufficiently unlikely.
- Ultimate limit state: a failure mechanism forms in the ground or wall, or severe structural damage occurs. States before collapse (e.g. forward sliding of a gravity wall) are also treated as ultimate.
- Serviceability limit state: movements or strains that make the wall unsightly, need unforeseen maintenance, shorten its life, or affect nearby structures and services.
- Checks can be direct (calculate that the limit state is not reached) or indirect (e.g. the middle-third rule for gravity walls as a deformation check).
- Walls should be ductile: give visible warning before failure, not collapse suddenly.

How are partial safety factors applied in Geoguide 1?

- Factored load $F_f = \gamma_f F$; factored material parameter $X_f = X / \gamma_m$.
- Typical ULS values: soil and wall dead weight $\gamma_f = 1.0$; surcharge $\gamma_f = 1.5$; soil strength c' and $\tan \phi'$ $\gamma_m = 1.2$. Check Tables 6 and 7 for the full set (e.g. undrained strength).

$$\tan \phi'_f = \tan \phi' / \gamma_m \quad c'_f = c' / \gamma_m$$

- Wall friction δ is then assessed from the factored ϕ'_f and ultimate earth pressures use the factored values.
- Sliding: factored base resistance from $\tan \delta_{bf} = (\tan \delta_b) / \gamma_m$, with δ_b from unfactored parameters.
- Serviceability checks: all factors 1.0, except surcharge with a favourable effect gets $\gamma_f = 0$.
- Water pressure: no factor, but use the worst credible groundwater condition.
- Structural design uses unfactored geotechnical loads, then the structural code's load factors.

Assessor follow-up

Why factor $\tan \phi'$ rather than ϕ' itself?

What is the 'worst credible groundwater condition' for retaining wall design?

- The groundwater condition that would arise in extreme events selected for design, assessed from catchment area, recharge potential and drainage.
- It is **not** the worst possible condition out of context, such as water up to the top of the wall regardless of drainage.
- It is used unfactored in ULS checks; the safety margin comes from the selection itself.
- Accidental water loads (e.g. a burst main) are hard to quantify and are best handled by prescriptive measures such as re-routing or ducting services.

What backfill would you specify behind a retaining wall, and what must be excluded?

- Ideal: free-draining, durable, high strength and stiffness, free of harmful matter. Sandy soils from weathered granite or volcanic rocks are usual in Hong Kong if they meet the grading and plasticity limits.
- Exclude: peat, vegetation and organic matter; dangerous or combustible material; metal, rubber, plastic; materials prone to volume change (marine mud, swelling clays, collapsible soils); soluble or chemically aggressive material.
- Clay backfill is not recommended (swelling, shrinkage, consolidation, and much larger movement needed to reach active state). Uniform silts are hard to compact. Weathered shales and mudstones, e.g. Lok Ma Chau Formation, can swell.
- Crushed rock fill is very suitable: hard, well graded, max particle 200 mm, not more than 30% decomposed, with a filter against the retained soil.
- Specify compliance tests: grading, Atterberg limits, moisture content, compaction.

What compaction standard does Geoguide 1 recommend for backfill?

- All backfill, filter and drainage material should be compacted, to **at least 95%** of maximum dry density.
- For level backfill, the **top 1.5 m** should be fine soil compacted to **at least 98%**, to limit infiltration, whether or not the surface is paved.
- Sloping backfill: the same standard for a vertical thickness of **at least 3 m**.
- Design strength must match the compaction specified. Heavy compaction plant should not be used close to the wall unless it is very massive; specify the allowable compaction plant.

Why can compaction cause higher pressure on a wall than classical earth pressure theory predicts?

- Compaction plant applies a large temporary load. When it passes, some horizontal stress is locked in, because soil does not unload elastically.
- Near the top, the locked-in lateral pressure can exceed active pressure. Below a certain depth, the normal active or at-rest pressure governs (Ingold's approach).
- Effect: higher pressure on the upper part of the wall, important for stems and light walls.
- Control: account for it in design, limit plant size near the wall, and state the allowable plant on drawings.

How does wall movement affect the earth pressure you design for?

- No movement: at-rest pressure (K_0).
Wall moving away from soil: pressure drops to active. Wall pushed into soil: pressure rises to passive.
- Only a small movement is needed for active pressure in granular soil; passive needs much more, so passive is usually only partly mobilised in service.
- Rigid or propped walls (e.g. basement walls, walls tied to structures) should be designed for at-rest or higher pressure.
- Cohesive backfill needs far more movement to reach active, so for a given movement the pressure is higher than for granular fill.
- The mode of movement (rotation about top or base, translation) changes the pressure distribution.

What factors govern the choice of retaining wall type?

- Location, nearby structures, space available and whether the support must stay within the site boundary.
- Wall height and the ground profile to be formed.
- Ground and groundwater conditions.
- Acceptable movement during construction and in service, and effects on neighbours.
- Material availability, construction time, appearance, design life and maintenance.
- Walls usually cost more than forming a slope, so first check whether a wall is needed, and keep the retained height low. Compare whole-life cost of alternatives.

How do you select soil shear strength parameters for retaining wall design (Geoguide 1 Ch. 5)?

- c' and ϕ' are coefficients of a simplified linear model, valid only over the stress range tested. Envelopes are usually curved; fit to the relevant stress range.
- Be careful with least-squares fits: too few results or variable soil can give a misleading c' .
- Lower bound guide: critical state ϕ'_{cv} about **34° for granitic** and **30° for volcanic** Hong Kong soils.
- Compacted fill and residual soil: a $c' = 0$ model is often used. Saprolite: consider relict joints and infill; use matrix strength unless corestones can be quantified.
- Watch sampling bias: tested specimens may be stronger (weak material breaks up) or weaker (coarse particles excluded) than the mass.
- Choose cautious "selected values" based on best estimates, test quality, variability and construction effects. Do sensitivity checks.

How much ground investigation and testing does a retaining wall need?

- Retained height under about 3 m: parameters can usually be based on previous tests on similar materials, with careful description and classification.
- Over about 6 m: determine parameters from laboratory tests on samples chosen by the designer, or from suitable field tests.
- In between: depends on how sensitive the wall dimensions are to the parameters. A preliminary design helps decide.
- Trial pits are recommended where possible, especially for foundations: they show the soil fabric and allow good block samples.
- Map exposures where persistent relict joints could affect the wall (e.g. bored pile walls), and review the design if adverse features appear during construction.

How would you estimate soil stiffness for retaining wall deformation in Hong Kong saprolite?

- Lab tests often underestimate stiffness because of sample disturbance. Back analysis of similar works is preferred.
- For granitic saprolite, effective Young's modulus has been correlated with SPT N: roughly **0.2N to 2N MN/m²** (Whiteside, 1986). The wide range reflects different derivation methods.
- Unloading or reloading stiffness is much higher: about **3 to 4 times** virgin loading for preliminary design.
- Stiffness depends on strain level, stress history, construction disturbance, and drainage.
- Field plate loading tests can be used for large walls, allowing for softening of the ground.

What is the Winkler (subgrade reaction) model, and what are its limitations for cantilevered walls?

- The wall is modelled as an elastic beam and the soil as a series of independent springs, with pressure proportional to deflection ($k_h = p/y$).
- For soil with stiffness increasing with depth, k_h is taken to increase linearly with depth.
- Limitations: grossly simplified; springs don't interact; the spring stiffness is strain-dependent, so values from very small movements are not valid at large strain.
- Do sensitivity analyses on spring values, and monitor wall deflection to check assumptions.

What checks do you carry out for a gravity retaining wall?

- ULS: sliding on the base (factored base friction), overturning, bearing capacity of the foundation, overall stability (slip surfaces passing below the wall), and structural failure of the wall body.
- SLS: limit deformation, commonly by the indirect middle-third rule (resultant within the middle third of the base, so no tension at the heel).
- Also check: foundation softening (blinding concrete on CDG below water), future trenches at the toe, drainage behind the wall.
- For crib and gabion walls: free-draining infill, durability of units or wire, and filters.

Assessor follow-up

Why does keeping the resultant in the middle third limit deformation?

What is the 'virtual back' of an L-shaped or inverted T retaining wall?

- A vertical plane through the heel. The soil between the stem and this plane is treated as part of the wall for stability checks.
- Earth pressure is calculated on the virtual back, and the weight of soil above the heel adds to stability.
- Surcharge between the stem and the virtual back helps stability, so it is given a factor of zero where favourable (serviceability), and treated cautiously in ULS checks.
- For structural design of the stem, calculate pressure on the actual stem.

How are cantilevered bored pile or caisson walls designed?

- Embedment from equilibrium of active pressure behind and passive resistance in front, using factored parameters for ULS.
- Where socketed into rock, a simplified wedge analysis is often used for socket design; consider joint orientation and mass strength.
- Persistent adverse relict joints or rock discontinuities can control behaviour, so do detailed geological mapping and review during construction.
- Deflection (SLS) by subgrade reaction or numerical analysis, with monitoring.
- Reduce passive resistance for possible future trenches at the toe.

How should a roadside retaining wall allow for future utility trenches?

- Geoguide 1: for roadside walls in built-up areas, it is prudent to assume a trench **at least 1 m deep** at the toe.
- Gravity and RC walls: set the foundation level so future trenching won't undermine the wall.
- Cantilevered walls: reduce passive resistance accordingly.
- Consult the relevant authorities on planned services.

How do you design drainage and filters behind a retaining wall?

- Provide a drainage layer behind the wall, protected by a filter, with weepholes or outlets. The design water pressure depends on how reliable the drainage is.
- Granular filters must meet retention and permeability criteria against the retained soil. Check grading of actual soil during construction. No-fines concrete must not be used as a filter.
- Geotextile filters are suitable in non-aggressive conditions (e.g. pH 5 to 10): use resistant polymers from established makers, protect from UV, and ensure good contact with the soil.
- Keep drainage clear by maintenance; blocked weepholes are a warning sign.

What surcharge loads must a retaining wall be designed for?

- Types: uniformly distributed, line, point, area, and horizontal loads, plus seismic.
- Geoguide 1 gives nominal surcharges for common situations; use these as minimums and add actual loads (buildings, cranes, stockpiles).
- Line and point loads: spread using elastic or other methods given in Geoguide 1.
- State limits on surcharge on the drawings, including during construction.
- Seismic load is generally not critical in Hong Kong (low to moderate seismicity), but follow Geoguide 1 Section 7.4.

What temporary works issues does Geoguide 1 raise for retaining wall construction?

- Responsibility for temporary works design must be clear, and contractor's designs controlled where risk is high.
- Design temporary works to the permanent standard, but allowing for shorter life and less severe conditions (e.g. groundwater).
- Temporary slopes: adequate stability, erosion protection, benching where over about 7.5 m high; remove weak material before backfilling.
- Protect foundations: CDG below water and weak rock deteriorate quickly, so place blinding immediately.
- Groundwater control to prevent piping and weakening; assess dewatering settlement on neighbours.
- Check partially completed walls (e.g. no backfill until top support is built).

When should permanent prestressed ground anchors be used in a retaining wall?

- Only when other methods are not practicable, because they impose a long-term maintenance commitment and recurring cost.
- Get the maintenance authority's or owner's agreement first.
- Assess installation suitability, effects on neighbouring structures and services, and rights of adjoining owners.
- Pay special attention to corrosion protection and creep. Follow Geospec 1.

What must be in place before using the observational method for a retaining wall?

1. Acceptable limits of behaviour established.
2. Range of possible behaviour assessed, showing it is likely to be within limits.
3. A monitoring plan that detects behaviour early enough, with fast enough instruments, warning levels, and experienced staff.
4. A contingency plan if limits are exceeded.

Typical use: walls supporting embankments on soft ground, monitoring movement and pore pressure.

Contingencies: regrading, extra support, drainage, or slowing construction.

What should a geotechnical design report for a retaining wall contain?

- Site description: topography, geology, groundwater and surface water, history, utilities and records.
- Critical interpretation of GI and monitoring results.
- Schedule of design assumptions: selected parameters, loads, geometry.
- Anticipated problems and contingency actions.
- Requirements for construction, testing, inspection and maintenance.
- Stability analyses and design calculations.
- Keep factual and interpreted information separate, and record construction-stage checks in an addendum.

How do you deal with relict joints and corestones when designing a wall in saprolite?

- Relict joints can cause failure along planes not assumed in soil mechanics theory. Inspect exposures, and map in detail where the wall could be affected (e.g. bored pile walls).
- Use a suitable calculation model and joint strength where adverse joints exist; infill (e.g. kaolin) can be much weaker.
- Corestones: conservative practice is to use matrix strength only, unless their proportion and distribution are reliably known.
- Map during construction and review the design if adverse features appear.

What rock parameters might you need for a retaining wall, and how are they estimated?

- Intact strength is rarely critical but is a useful index; Grade III rock is very variable and should be taken conservatively.
- Joint shear strength: basic friction plus roughness at low stress ($\phi_b + i$), or Barton's model; infilled joints can be very weak.
- Rock mass strength: Hoek-Brown for jointed rock that can be treated as isotropic; otherwise wedge or distinct element analysis.
- Rock mass modulus: much lower than intact; e.g. via a mass factor (BS 8004), modulus ratio (200 to 500 for most Hong Kong rocks), or RMR correlation.

What maintenance does a retaining wall need, and how should design allow for it?

- Routine Maintenance Inspections (clear weepholes and drains, remove vegetation, repair cracks) and periodic Engineer Inspections (Geoguide 5).
- Design for maintenance: access, durable materials, drainage that can be cleared, and avoiding features that are hard to inspect.
- Identify maintenance items in the design report and maintenance manual.
- Watch for warning signs: bulging, cracking, blocked weepholes, seepage, leaking services, trees on the wall.